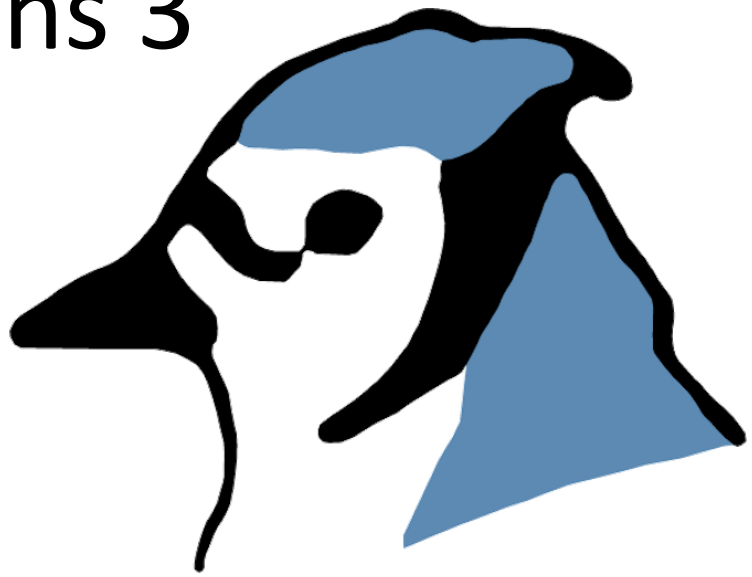
Understanding class definitions 3

Exploring source code



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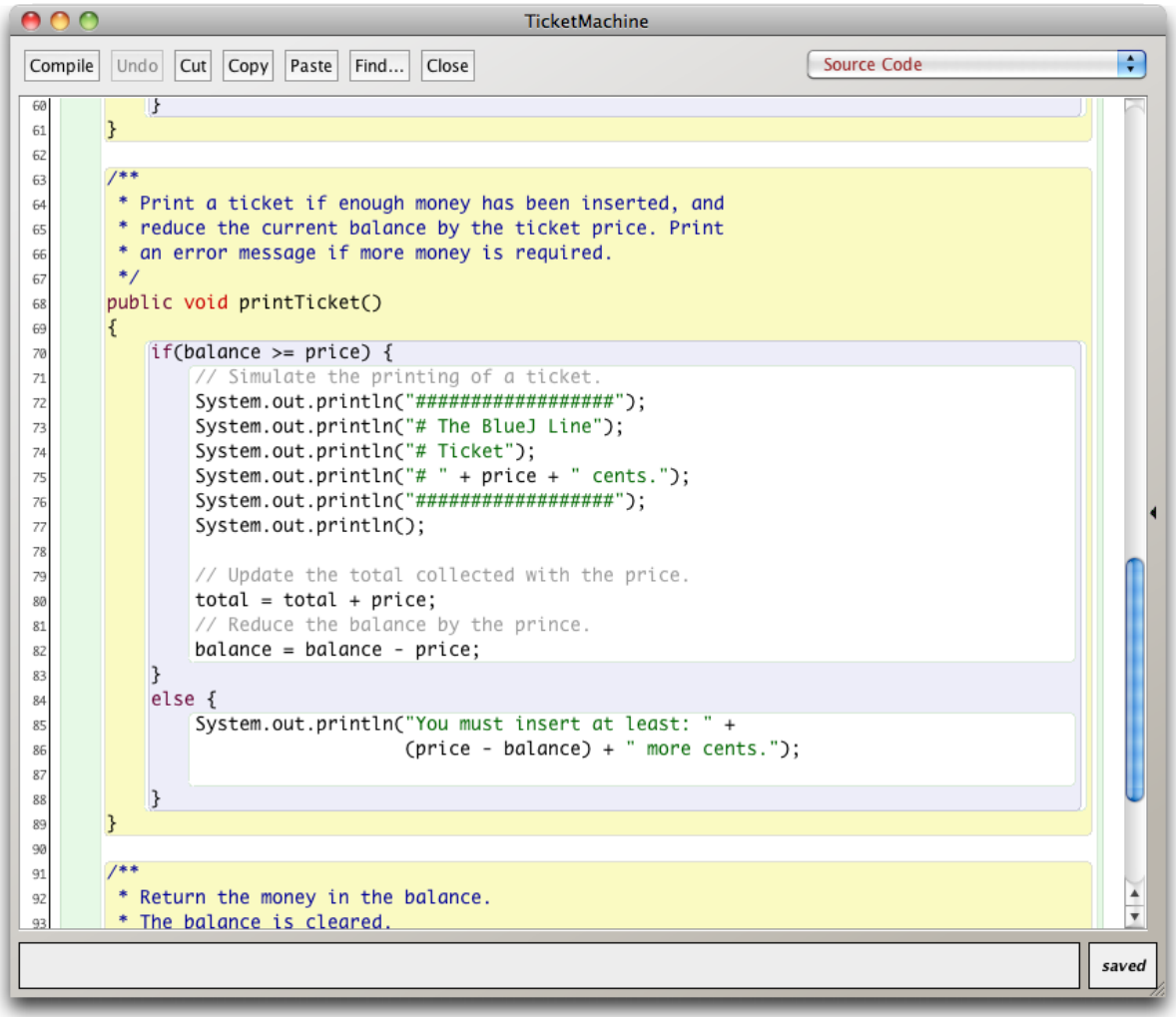
Upcoming

- Local variables.

Variables – a recap

- Fields are one sort of variable.
  - They store values through the life of an object.
  - They are accessible throughout the class.
- Parameters are another sort of variable:
  - They receive values from outside the method.
  - They help a method complete its task.
  - Each call to the method receives a fresh set of values.
  - Parameter values are short lived.

Scope highlighting



Scope and lifetime

- Each block defines a new scope.
  - Class, method and statement.
- Scopes may be nested:
  - statement block inside another block inside a method body inside a class body.
- Scope is static (textual).
- Lifetime is dynamic (runtime).

To think about

- How could we write a method to ‘refund’ an excess balance?
  - We need to return the value in balance.
  - We need to set the balance to zero.
- How can we do the first without losing the value in balance?

Unsuccessful attempt

```
public int refundBalance()
{
    // Return the amount left.
    return balance;
    // Clear the balance.
    balance = 0;
}
```

It looks logical, but the language does not allow it.

Local variables

- Methods can define their own, *local* variables:
  - Short lived, like parameters.
  - The method sets their values – unlike parameters, they do not receive external values.
  - Used for ‘temporary’ calculation and storage.
  - They exist only as long as the method is being executed.
  - They are only accessible from within the method.
  - They are defined within a particular *scope*.

Local variables

A local variable


```
public int refundBalance()
{
    int amountToRefund;
    amountToRefund = balance;
    balance = 0;
    return amountToRefund;
}
```

Scope and lifetime

- The scope of a field is its whole class.
- The lifetime of a field is the lifetime of its containing object.
- The scope of a local variable is the block in which it is declared.
- The lifetime of a local variable is the time of execution of the block in which it is declared.

Review (1)

- Class bodies contain fields, constructors and methods.
- Fields store values that determine an object's state.
- Constructors initialise objects – particularly their fields.
- Methods implement the behavior of objects.

Review (2)

- Fields, parameters and local variables are all variables.
- Fields persist for the lifetime of an object.
- Local variables are used for short-lived temporary storage.
- Parameters are used to receive values into a constructor or method.

Review (3)

- Methods have a return type.
- **`void`** methods do not return anything.
- non-**`void`** methods always return a value.
- non-**`void`** methods must have a return statement.

Review (4)

- ‘Correct’ behavior often requires objects to make decisions.
- Objects can make decisions via conditional (if) statements.
- A true-or-false test allows one of two alternative courses of actions to be taken.

Questions?